StoryDiffusion

Team Members

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Goal

What is the main goal of this work?

- Compare the results of vanilla SDXL and StoryDiffusion, and analyze the effectiveness of StoryDiffusion in maintaining character & scene consistency
- Identify the **optimal configurations** of the sa32 and sa64 parameters while systematically comparing the performance differences between **simplified prompts** and **highly narrative prompt**s in producing consistent, high-quality generations.
- Explore the extent of StoryDiffusion’s capabilities, and identify the model’s limitations

Main Reference Work

Core Problem:

- Existing diffusion models struggle to keep characters consistent across multiple images or video frames
- Reference-image methods reduce text controllability; video modules require heavy compute

Key Idea:

- Introduces Consistent Self-Attention (CSA): a training-free modification that forces cross-image consistency
- Adds a Semantic Motion Predictor (SMP): predicts transitions in semantic space for smoother videos

Main Reference Work

How It Works:

- Split story into multiple prompts; generate all frames in a batch
- CSA mixes tokens across images → consistent identity, attire, and style
- SMP converts frames into long, stable transition videos

Results:

- Strong user preference over IP-Adapter, PhotoMaker, SEINE, SparseCtrl
- Zero training, plug-and-play, maintains high prompt controllability
- Produces long, stable image sequences and videos

Pipeline

- SDXL instead of default RealVision
- Customized outputs:
  - Style – Comic
  - No captions, no comic-style layout
- pipe = StableDiffusionXLPipeline.from_pretrained(
  sd_model_path, torch_dtype=torch.float16,
  use_safetensors=True, variant="fp16")
  - use_safetensors=True
  - variant=”fp16”

SDXL results compared with StoryDiffusion

Goal:

Compare results of vanilla SDXL & StoryDiffusion, analyze effectiveness of StoryDiffusion in maintaining character & scene consistency

Pipeline:

Create story prompts ➞ SDXL generation ➞ StoryDiffusion generation (CSA) ➞ compare

Experimental Setup:

Controlled: same prompts, same character, same number of frames

Varied: generation method (SDXL vs StoryDiffusion)

Evaluation dimensions: character identity, attire consistency, scenic/stylistic consistency, prompt adherence

Use of AI:

Helpful for: conceptual explanations, debugging runtime errors

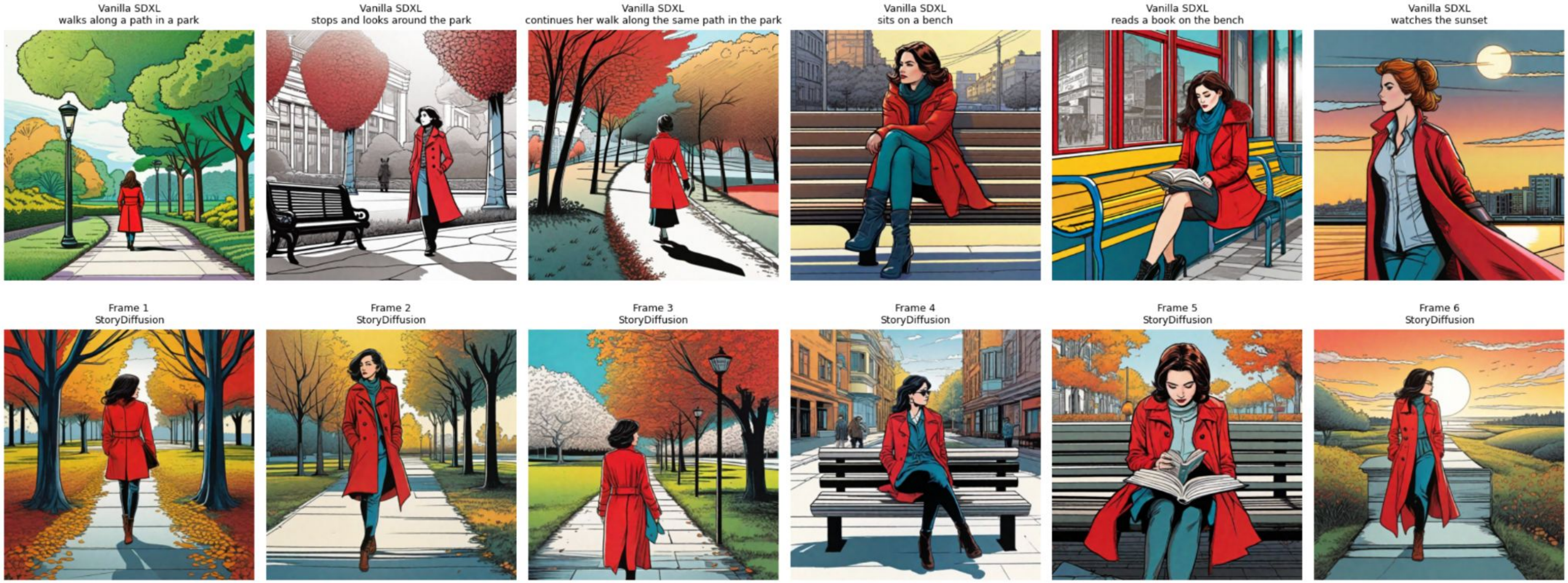
Unhelpful for: debugging OOM errors, prompt engineering, interpreting results

SDXL results compared with StoryDiffusion

Story prompts:

```
general_prompt = "a woman with a red coat"
prompt_array = ["walks along a path in a park",
               "stops and looks around the park",
               "continues her walk along the same path in the park",
               "sits on a bench",
               "reads a book on the bench",
               "watches the sunset"]
```

a woman with a red coat :



Vanilla SDXL Frame 5

Vanilla SDXL Frame 6

StoryDiffusion Frame 4

StoryDiffusion Frame 3

StoryDiffusion Frame 5

SDXL results compared with StoryDiffusion

Story prompts:

```
general_prompt = "a woman with glasses and a brown sweater"
prompt_array = ["walking through a busy city",
               "shopping at the mall",
               "sits on a park bench",
               "eats lunch at a restaurant",
               "visits a museum",
               "watches the sunset in the park"]
```

SDXL results compared with StoryDiffusion

Summary of Findings: effectiveness of StoryDiffusion in maintaining…

1. **Character consistency**
   - Significantly better identity consistency
   - Attire consistency slightly improved; highly prompt-dependent
2. **Scene consistency**
   - Stronger scene and stylistic consistency
   - But character consistency > scene consistency
3. **Prompt adherence**
   - Differences primarily determined by prompt engineering, not the model

Potential observed training bias

- Recurring clothing styles, colors, and unexpected elements suggest underlying dataset biases

sa32 and sa64 Optimization: Experimental Setup

Goal: Identify the **optimal configurations** of the sa32 and sa64 parameters while systematically comparing the performance differences between **simplified prompts** and **highly narrative prompts** in producing consistent, high-quality generations.

- **sa32:** responsible for the broad structure, layout, and composition of the image.
- **sa64:** responsible for fine details, textures, and specific features.
- **Quality of prompts:** Narrative vs Simplified prompts

sa32 and sa64 Optimization: Experimental Setup

Evaluation Metric:

- CLIP Score: measures the compatibility of image-caption pairs
- Visual evaluation

Methodology:

- Run a grid search on sa32 and sa64 from **0.2** to **0.8** with increment of 0.1
- Select the best performing hyperparameter setting
- Run parallel generations using both <u>detailed, narrative prompts</u> and <u>simplified descriptions</u>, and comparing the results visually.

sa32 and sa64 Optimization: Experimental Results

Story Prompts:

```
general_prompt = "a boy in pink shorts with a buzz cut, high quality, 8k"
negative_prompt = "naked, deformed, bad anatomy, disfigured, poorly drawn face, mutation, extra limb, ugly, disgusting, poorly drawn hands,
prompt_array = [
    "eating pancakes with his father at a kitchen table",
    "riding a bicycle wearing a helmet on a street",
    "playing basketball outdoors with a blonde male friend",
    "reaching for a book on a shelf in a library",
    "eating fried chicken with his mother at a dinner table"
]
```

Results (Top 5):

```
========================================
Grid Search Results
========================================
 Guidance  sa32 (Pose)  sa64 (Face)  CLIP Score
      5.0          0.4          0.4       39.18
      5.0          0.4          0.5       39.08
      5.0          0.4          0.3       39.07
      5.0          0.4          0.2       39.04
      5.0          0.4          0.6       39.03
```

sa32 and sa64 Optimization: Experimental Results

Simplified

```
general_prompt = "a boy in pink shorts with a buzz cut, high quality, 8k"
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prompt_array = [
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    "playing basketball outdoors with a blonde male friend",
    "reaching for a book on a shelf in a library",
    "eating fried chicken with his mother at a dinner table"
]
```

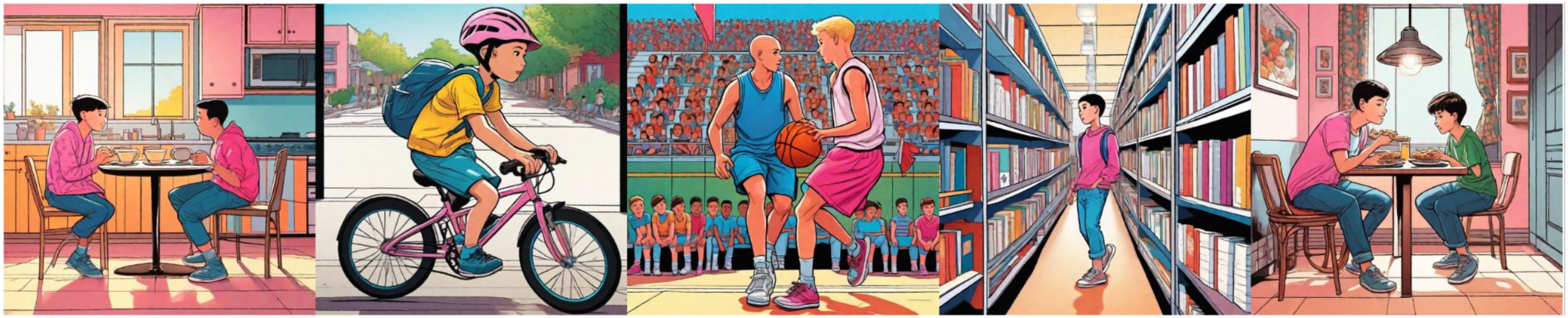
Narrative

```
general_prompt = "a boy in pink shorts with a buzz cut, high quality, 8k, detailed features"
negative_prompt = "naked, deformed, bad anatomy, disfigured, poorly drawn face, mutation, extra limb, ugly, disgusting, poorly drawn hands, missing li
prompt_array = [
    "Morning atmosphere, sitting at a sunny kitchen table eating fluffy pancakes, laughing and talking with his father, warm sunlight streaming throug
    "Dynamic action shot, riding a bicycle energetically to school while wearing a safety helmet, wind blowing, motion blur, suburban street backgroun
    "Daytime at school, playing a competitive game of basketball on the outdoor court, guarding a blonde male friend, athletic pose, blue sky, energet
    "Quiet atmosphere, walking through the library aisles and reaching out to pick a hardcover book from a high shelf, surrounded by rows of books, st
    "Night time, sitting at the dinner table eating crispy fried chicken with his mother, warm lamp lighting, dark window background, intimate family
]
```

sa32 and sa64 Optimization: Experimental Results

Simplified

Narrative



sa32 and sa64 Optimization: Key Takeaways

- **Insight 1**: Running a quick gridsearch allows us to systematically find the most optimal hyperparameter for sa32 and sa64. However, we must recognise that this is entirely dependent on the CLIP Score derived mathematically and may not always be accurate.

- CLIP Score is a reference-free evaluation metric that measures how well a generated caption aligns with the visual content of an image, and the similarity between textual or visual inputs. It has been proven to correlate strongly with human judgments.

$$\text{CLIPScore(I, C)} = max(100 * cos(E_I, E_C), 0)$$

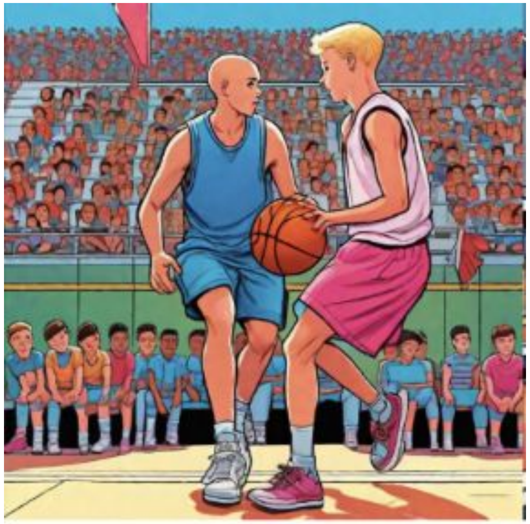
where *Ei* and *Ec* are the embeddings for image and caption respectively

CLIP score¶. CLIP Score - PyTorch-Metrics 1.8.2 documentation. (n.d.). https://lightning.ai/docs/torchmetrics/stable/multimodal/clip_score.html

sa32 and sa64 Optimization: Key Takeaways

- **Insight 2:** The model runs better with more narrative prompts as shorter prompts provides little information for image generation. This is evident from the 3rd frame where the basketball game for narrative prompt is much better generated than the simplified prompt. Moreover, the simplified prompt produces inconsistent hair type of the boy as he went from having a buzz cut to being bald from after riding a bicycle. On the other hand, narrative prompts manages to keep the hair type of the boy consistent.

Simplified



Narrative

sa32 and sa64 Optimization: Key Takeaways

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Simplified

sa32 and sa64 Optimization: Key Takeaways

- **Insight 3:** The model fails to handle multiple characters as the introduction of "father", "mother" and "blonde male friend" confuses the model into merging the features of pink shorts and boy with buzz cut with them. Moreover, the definition of the boy in terms of him being in pink seems to carry over to the other characters and hence results in poorly generated images of "father" and "mother".

- **Insight 4:** The model misinterprets shorts and shirts as most of the image generated produces the boy in pink shirt instead of shorts. This is likely due to how similar the words are and highlights a weakness of the model

Simplified

Narrative

StoryDiffusion Narratives (Diane)

- Goal: Use StoryDiffusion to recreate narratives from well-known animated children's movies, in order to probe where the model is accurate, where it fails, and why.
- Methodology:
  - Systematically experiment with different prompt styles to identify what qualities make some prompts more effective than others.

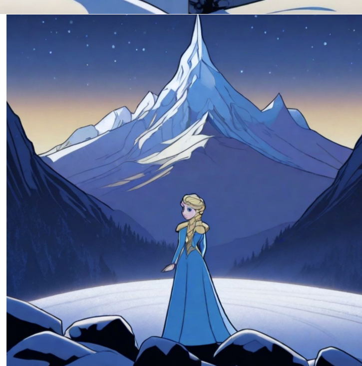
Key Takeaway (1)

- The first tokens are taken as the most important.
  - Second prompt set is focused on camera work, while the first has more narrative elements.

```
prompt_array = [
    "Scene 1: Elsa stands alone on a snowy peak at night, looking down as wind blows her cape and snow falls.",
    "Scene 2: From a side angle, Elsa sends a surge of blue ice forward, creating the first shimmering steps.",
    "Scene 3: A wider shot reveals Elsa climbing the newly formed staircase, frost spiraling in the air.",
    "Scene 4: A distant view shows the circular ice platform blooming outward as Elsa stands atop it.",
    "Scene 5: At dawn, the completed crystalline palace rises high against a purple sky, glowing in soft light."
]
```

```
prompt_array = [
    # 1 — standing on empty slope before magic
    "Scene 1: Wide shot; open snow field with Elsa alone at center-left, mountains behind; then add falling snow and her tense posture.",

    # 2 — first ice burst forming structural arc
    "Scene 2: Side wide shot; arc of ice forming from Elsa toward screen right; then add glowing frost particles and blue shimmer.",

    # 3 — staircase creation with clear layout
    "Scene 3: Low-angle shot; rising diagonal staircase of ice leading upward from lower-left to upper-right; then add her stepping forward.",

    # 4 — circular platform forming mid-air
    "Scene 4: Distant wide shot; floating circular ice platform centered above slope; then add ornate fractal swirls and sparkling trails.",

    # 5 — final palace tower against dawn
    "Scene 5: Extreme wide shot; tall crystalline palace occupying right side of frame, dawn gradient sky behind; then add reflective facets."
]
```



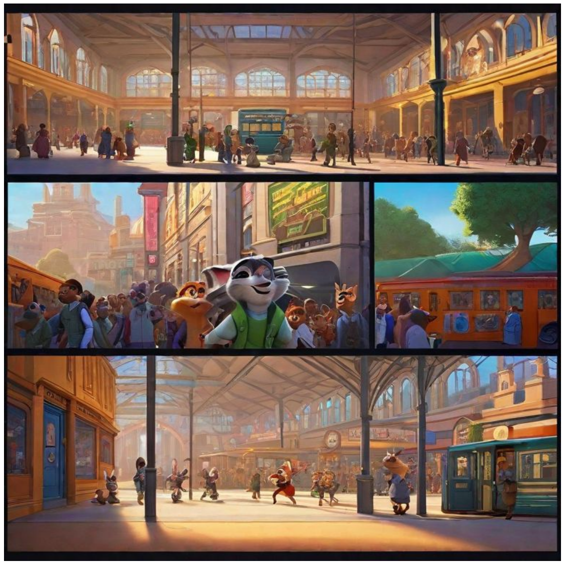




Key Takeaway (2)

- Consistent prompt skeleton gives better continuity, but also risks overgeneralization.
  - Reuse a simple template for every frame:
    - Ex) Shot type + camera angle → setting → characters + positions → action → mood → style

```
prompt_array = [
    "Scene 1: Wide shot of the train gliding into the station, Judy looking out the window at the sprawling Zootopia skyline.",
    "Scene 2: Medium shot as Judy steps off the train platform, surrounded by tall animals rushing by.",
    "Scene 3: Low-angle shot showing Judy dwarfed by towering station architecture and bustling crowds.",
    "Scene 4: Over-the-shoulder shot as Judy walks toward the massive station exit, sunlight pouring in.",
    "Scene 5: Wide exterior shot of Judy outside the station, gazing up at the city’s towering districts."
]
```

```
prompt_array = [
    # Shot type + angle → setting → characters + positions → action → mood → style
    "Wide frontal shot; bright station interior; Judy centered at the platform edge; she looks out at the skyline; hopeful mood; Pixar-style clarity.",
    "Medium forward-facing shot; same station hall; Judy centered as crowds move behind her; she steps off the train; excited mood; clean Pixar lighting."
    "Low-angle steady shot; towering glass architecture; Judy small at center foreground; she pauses, taking in the space; awe-filled mood; stylized soft
    "Over-the-shoulder consistent angle; station exit glowing ahead; Judy left-of-center walking toward the doors; determined mood; crisp Pixar-style rend
    "Wide exterior shot; same forward orientation; Judy centered facing the city's districts; she breathes in the moment; inspired mood; bright animated h
]
```

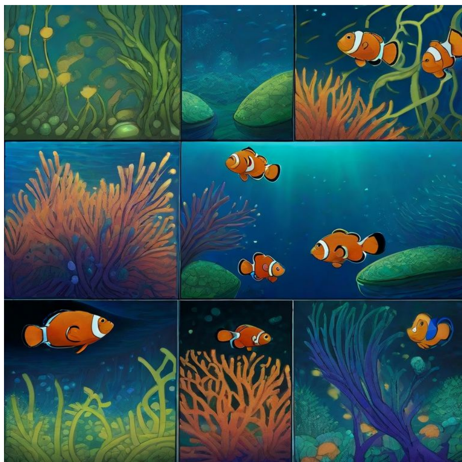
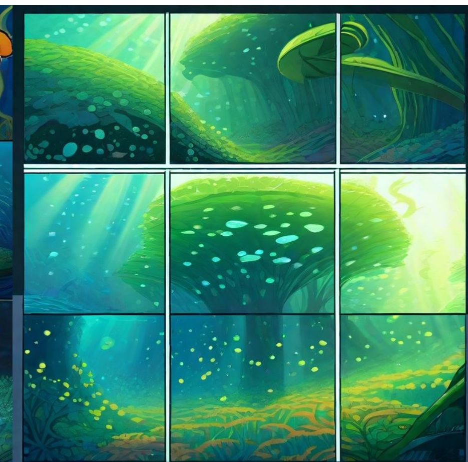
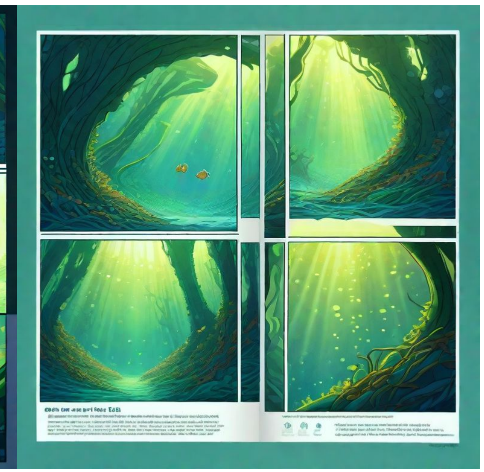
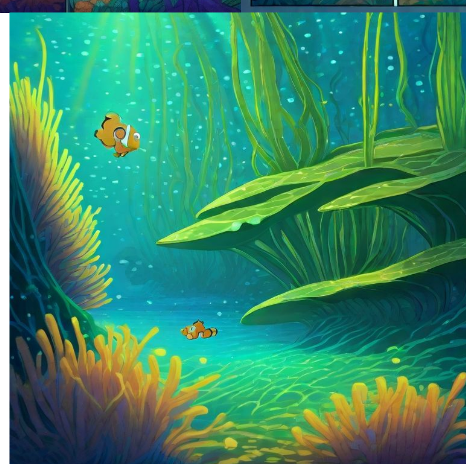
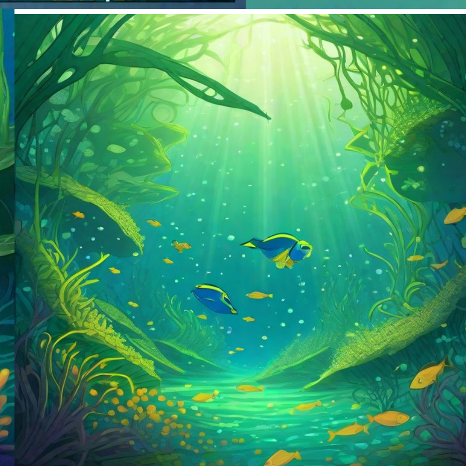


Key Takeaways (3)

- Comparing minimal narrative prompts v. verbose narrative prompts
  - The model can only process ~77 tokens at a time

```
prompt_array = [
    "Scene 1: Nemo swims through kelp; Dory follows.",
    "Scene 2: Nemo turns slightly; Dory moves closer.",
    "Scene 3: Both fish head toward a brighter area.",
    "Scene 4: Nemo pauses near a kelp strand; Dory hovers.",
    "Scene 5: Nemo and Dory enter a wider clearing."
]
```

```
prompt_array = [
    "Scene 1: Nemo, a small vibrant orange clownfish with a lucky fin, guides the path through dense swaying kelp as Dory, a tall blue tang with"
    "wide expressive eyes, follows amid drifting particles, rippling caustic light, soft underwater shadows, and richly textured fronds surrounding them.",

    "Scene 2: From a side view, Nemo threads between tall overlapping kelp ribbons while Dory arcs closer, her bright fins catching shifting"
    "highlights as glowing particles drift upward, the forest shimmering with layered greens, subtle motion, and cinematic Pixar-style underwater depth.",

    "Scene 3: Overhead perspective reveals Nemo descending along a winding kelp corridor as Dory glides beneath him, surrounded by luminous drifting"
    "debris, bending kelp silhouettes, gently pulsing caustic reflections, and warm shafts of filtered sunlight cutting through the deep blue water.",

    "Scene 4: Nemo pauses beside a softly glowing kelp frond while Dory hovers nearby, both illuminated by shimmering highlights, swirling particles,"
    "and interlaced shadows as the forest shifts gently around them, creating layered depth, warm color vibrance, and expressive cinematic detail.",

    "Scene 5: The two fish swim into a widening clearing where kelp thins, sunlight pours downward, drifting particles glitter, fronds sway in rhythmic"
    "motion, and Nemo and Dory move together through radiant water filled with caustic light, layered reflections, and richly animated underwater color."
    ]
```

Key Takeaway (4)

- StoryDiffusion struggles with fantasy/magical elements – it lacks true “imagination”
  - Tends to skip confusing elements entirely

```
# 3 — staircase creation with clear layout
"Scene 3: Low-angle shot; rising diagonal staircase of ice leading upward from lower-left to upper-right; then add her stepping forward.",
```

Limitations

- Struggles to capture precise spatial relationships and complex compositions
  - Often gives some but not all elements, or rearranges them.
- Fragile continuity
  - Character poses, positions, or expressions jump unpredictably,
  - Props may vanish or reappear
  - Backgrounds subtly change style or layout
- Shallow causal/narrative logic
  - The scenes are independent snapshots loosely tied by text, rather than a deeply understood storyboard

Thank you!

QUESTIONS?